

Disproof of conjecture 00000001044

earthking11

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Verdict: FALSE

We disprove conjecture 00000001044 as stated. The refutation is unconditional, elementary, and already complete at the smallest non-trivial witness $(q, n) = (3, 2)$: the hypothesis $d = \gcd(n, q^2 - 1) > 1$ holds and $(q - 1)/d = 1$, yet *every* $a \in \mathbb{F}_3$ gives a value set whose complement has size 1. Hence $a = 0$ is not the unique minimiser, and the other a do not give a strictly larger value set — they tie it.

1 The conjecture, quoted

Quoted verbatim from `conjectures/00000001044.md`:

English. Definition: The value set of the Dickson polynomial $D_n(x, a)$ is its image over $\mathbb{F}_q$. Conjecture: When $\gcd(n, q^2 - 1) = d > 1$ (the non-permutation case), the minimum size of the complement of the value set is $(q - 1)/d$, attained at $a = 0$ (the monomial x^n); every other a gives a strictly larger value set.

中文。定义: Dickson 多项式 $D_n(x, a)$ 的值集指其在 $\mathbb{F}_q$ 上的像。猜想: 当 $\gcd(n, q^2 - 1) = d > 1$ (非置换情形) 时, 值集补集的最小尺寸为 $(q - 1)/d$, 且该极值在 $a = 0$ (单项式 x^n) 时取得; 一切其他 a 给出严格更大的值集。

The claim has two clauses:

(C1) the minimum complement size equals $(q - 1)/d$ and is attained at $a = 0$;

(C2) every other a gives a *strictly larger* value set.

Both fail at $(q, n) = (3, 2)$; indeed (C1) and (C2) are mutually inconsistent as soon as any other a ties $a = 0$ (Section 4).

2 Setup

Definition 1 (Dickson polynomial). For a field F and $a \in F$, the Dickson polynomial $D_n(x, a) \in F[x]$ is defined by

$$D_0(x, a) = 2, \quad D_1(x, a) = x, \quad D_k(x, a) = x D_{k-1}(x, a) - a D_{k-2}(x, a) \quad (k \geq 2).$$

In particular $D_2(x, a) = x^2 - 2a$ and $D_3(x, a) = x^3 - 3ax$.

Definition 2 (Value set and complement). The *value set* of $D_n(\cdot, a)$ over $\mathbb{F}_q$ is $V_q(n, a) = \{ D_n(x, a) : x \in \mathbb{F}_q \} \subseteq \mathbb{F}_q$, and its *complement size* is $\text{comp}_q(n, a) = q - |V_q(n, a)|$.

Fix n and q , and let $d = \gcd(n, q^2 - 1)$. The conjecture concerns the regime $d > 1$.

3 Primary refutation: $(q, n) = (3, 2)$

Take $q = 3$ and $n = 2$, so $D_2(x, a) = x^2 - 2a$ over $\mathbb{F}_3$. Then

$$d = \gcd(n, q^2 - 1) = \gcd(2, 9 - 1) = \gcd(2, 8) = 2 > 1, \quad \frac{q-1}{d} = \frac{3-1}{2} = 1.$$

So the hypothesis of the conjecture holds strictly.

Theorem 3. For $(q, n) = (3, 2)$, every $a \in \mathbb{F}_3$ gives a value set of size 2:

$$V_3(2, 0) = \{0, 1\}, \quad V_3(2, 1) = \{1, 2\}, \quad V_3(2, 2) = \{0, 2\},$$

hence

$$\text{comp}_3(2, 0) = \text{comp}_3(2, 1) = \text{comp}_3(2, 2) = 1.$$

In particular $a = 0$ is not the unique minimiser, and neither $a = 1$ nor $a = 2$ gives a strictly larger value set than $a = 0$.

证明. Direct evaluation over $\mathbb{F}_3$ of $D_2(x, a) = x^2 - 2a$:

a	$x \mapsto D_2(x, a)$ on $x = 0, 1, 2$	value set	complement
0	0, 1, 1	$\{0, 1\}$	1
1	1, 2, 2	$\{1, 2\}$	1
2	2, 0, 0	$\{0, 2\}$	1

Each value set has size 2, so each complement has size $3 - 2 = 1$. All three complement sizes are equal, and the value sets all have the same size, so none is strictly larger than another. $\square$

Corollary 4. Conjecture 000000001044 is false. At $(q, n) = (3, 2)$ the hypothesis $d = 2 > 1$ holds, the conjectured value $(q - 1)/d = 1$ is correct as a minimum, but it is attained by all three choices $a = 0, 1, 2$, and no value set is strictly larger than that of $a = 0$.

4 The statement is internally inconsistent

Independently of any computation, clauses (C1) and (C2) contradict each other. Write $v(a) = |V_q(n, a)|$ for the value-set size, so that the complement size is $q - v(a)$. Clause (C1) says that $a = 0$ *minimises the complement*, i.e. $q - v(0) \leq q - v(a)$, equivalently $v(a) \leq v(0)$, for every a . Clause (C2) says that every $a \neq 0$ has a *strictly larger value set*, i.e. $v(a) > v(0)$. The two requirements $v(a) \leq v(0)$ and $v(a) > v(0)$ cannot both hold for any $a \neq 0$. Thus, whenever $\mathbb{F}_q$ contains more than one element (so that some $a \neq 0$ exists), the conjecture as written is unsatisfiable: a minimiser of the complement whose other points all have strictly larger value sets cannot exist. The plausible intended clause (C2) would have been “every other a gives a strictly *smaller* value set” (i.e. a strictly larger complement), which is also what the examples below refute.

At $(q, n) = (3, 2)$ the inconsistency is visible concretely: $a = 1$ has $v(1) = v(0) = 2$, violating the strict inequality $v(1) > v(0)$ demanded by (C2), while also realising the minimum demanded by (C1).

5 The same tie at $(5, 2)$ and $(7, 2)$

The primary witness is not an isolated accident. For $n = 2$ we have $D_2(x, a) = x^2 - 2a$, and the value set is the set of squares shifted by $-2a$; when 2 is invertible in $\mathbb{F}_q$, as a ranges over $\mathbb{F}_q$ the shift ranges over all of $\mathbb{F}_q$, so every a produces a translate of the same square set.

(q, n)	$d = \gcd(n, q^2 - 1)$	$(q - 1)/d$	complement sizes	$\text{comp}_q(n, a)$	value-set sizes
$(5, 2)$	$\gcd(2, 24) = 2$	2	$a = 0, 1, 2, 3, 4 :$	2, 2, 2, 2, 2	all 3
$(7, 2)$	$\gcd(2, 48) = 2$	3	$a = 0, \dots, 6 :$	3, 3, 3, 3, 3, 3, 3	all 4

In both rows the hypothesis $d > 1$ and the numerical value $(q - 1)/d$ are correct, but the minimum is attained by *every* a , not uniquely by $a = 0$, and no a yields a strictly larger value set. Thus clause (C2) fails across an infinite family of $n = 2$ examples, not merely at $q = 3$.

6 The reversal at $(7, 3)$: $a = 0$ is the worst choice

The direction of the claim also fails. Take $q = 7$, $n = 3$, so $D_3(x, a) = x^3 - 3ax$ over $\mathbb{F}_7$. Then

$$d = \gcd(3, 49 - 1) = \gcd(3, 48) = 3 > 1, \quad \frac{q - 1}{d} = \frac{6}{3} = 2.$$

a	value set $V_7(3, a)$	$ V_7(3, a) $	complement
0	$\{0, 1, 6\}$ (the cubes)	3	4
1	$\{0, 2, 3, 4, 5\}$	5	2
2	$\{0, 2, 3, 4, 5\}$	5	2
3	$\{0, 1, 3, 4, 6\}$	5	2
4	$\{0, 2, 3, 4, 5\}$	5	2
5	$\{0, 1, 3, 4, 6\}$	5	2
6	$\{0, 1, 3, 4, 6\}$	5	2

Here $a = 0$ gives the *largest* complement, 4, which strictly exceeds the conjectured minimum $(q - 1)/d = 2$, while every nonzero a gives the smaller complement 2. So $a = 0$ maximises the complement instead of minimising it: the conjecture has the direction exactly backwards at $(q, n) = (7, 3)$.

7 Further breakages of the formula and of the “non-permutation” label

Two further defects of the statement are worth recording.

- $(q, n) = (5, 3)$. Here $d = \gcd(3, 24) = 3 > 1$, but d does not divide $q - 1 = 4$, so $(q - 1)/d = 4/3$ is not an integer and cannot be a set size. Moreover $D_3(x, 0) = x^3$ permutes $\mathbb{F}_5$ (as $\gcd(3, 5 - 1) = 1$), so $V_5(3, 0) = \mathbb{F}_5$ and $\text{comp}_5(3, 0) = 0$. Thus in the very case the conjecture labels “non-permutation” ($d > 1$), the monomial $a = 0$ is in fact a permutation polynomial, and the true minimum complement is 0, not $(q - 1)/d$.
- $(q, n) = (5, 4)$. Here $d = \gcd(4, 24) = 4$ and $(q - 1)/d = 1$, but the true minimum complement is 2 (attained at $a = 2, 3$), and no $a \in \mathbb{F}_5$ attains the claimed value 1 at all: $\text{comp}_5(4, a) = 3, 3, 2, 2, 3$ for $a = 0, \dots, 4$, with $\text{comp}_5(4, 0) = 3$. So even the numerical value $(q - 1)/d$ can fail outright.

8 The torus-reading caveat

One might try to rescue the claim by changing the reading of “value set” from the image over $\mathbb{F}_q$ to the image on the norm-1 torus, i.e. the subgroup of order $q + 1$ of $\mathbb{F}_{q^2}^\times$, rather than on $\mathbb{F}_q$. Under that reading the monomial $a = 0$ *is* the unique minimiser, so the “uniqueness at $a = 0$ ” half of the claim survives in a qualified sense. But the minimum complement there is 0, not $(q - 1)/d$; the value $(q - 1)/d$ is not what the torus reading produces. Consequently that reading does not save the stated formula either, and in any case it contradicts the conjecture’s own definition, which

explicitly places the value set over $\mathbb{F}_q$. The refutation above uses the definition as written.

9 Reproducibility

The brute-force checks are carried out by `reproduce.py`, which uses the Python 3 standard library only:

```
$ python3 reproduce.py
... (tables for (3,2), (5,2), (7,2), (7,3), (5,3), (5,4))
PASS: all checks verified; conjecture 00000001044 is FALSE.
```

It implements the Dickson recursion over the residues of $\mathbb{F}_q$, computes the complement size for every a , and asserts the (3,2) tie, the (5,2) and (7,2) ties, and the (7,3) reversal; it prints PASS and exits 0 exactly when every check holds.

The same facts are formalised in the Lean 4 project under `lean4/`, using core Lean only (no Mathlib) and no `sorry`. Its main statements are

```
theorem conjecture_00000001044_false : ...
theorem q3n2_tie : tieWithZero 3 2 = true
theorem q7n3_a0_largest :  $\forall a, a < 7 \rightarrow \text{compSize } 7 \ 3 \ a \leq \text{compSize } 7 \ 3 \ 0$ 
theorem q5n2_all :  $\forall a, a < 5 \rightarrow \text{compSize } 5 \ 2 \ a = 2$ 
theorem q7n2_all :  $\forall a, a < 7 \rightarrow \text{compSize } 7 \ 2 \ a = 3$ 
```

where `D`, `valueSet` and `compSize` are the Dickson recursion, the value set, and its complement size on residues represented as naturals reduced modulo q . `lake build` compiles the project, and `lake env lean Check.lean` reports the axioms of every theorem: the concrete computations depend on no axioms at all (the two divisibility facts use only `propext`), and none reports `sorryAx`. See `lean4/README.md` for the full statement table.

10 Files

File	Description
<code>README.md</code>	Verdict, quoted conjecture, tables, caveats, file list, reproduction, status.
<code>main.tex</code>	This LaTeX source (standalone <code>article</code> , <code>tectonic main.tex</code>).
<code>build/main.pdf</code>	PDF produced by <code>tectonic</code> .
<code>reproduce.py</code>	Python 3 standard-library brute force; PASS/FAIL, non-zero exit on failure.
<code>lean4/Main.lean</code>	Lean 4 formalisation, core Lean only.
<code>lean4/Check.lean</code>	<code>#print axioms</code> audit.
<code>lean4/README.md</code>	Statement table and scope note.